SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science & Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code	**CSA4315**	**Course Title**	**Internet Programming**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 1: Application Based Questions**
Weightage	**40% of CIA**	**Total Marks**	**100**
CO1 Statement	*Explain the fundamentals of web architecture, client-server communication, Internet protocols, HTTP request and response messages, and configure LAMP/WAMP environments for web application development and deployment.* **(BL3)**		
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Code of Conduct:

I,DN SHIRSHIKA certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that anyviolation of academic integrity rules will result in disciplinary action.

Signature of the student:

CHARANI.A

8. A college website must handle 500 students accessing it simultaneously. Explain how the web server and database server should cooperate to process requests efficiently.

1. Introduction

When 500 students access a college website at the same time, the website must process all requests quickly without becoming slow or crashing.

For example, students may simultaneously check examination results, download study materials, view attendance details, register for courses, pay college fees, and check timetables or announcements.

To handle these requests efficiently, the web server and database server must work together using suitable techniques such as load balancing, caching, connection pooling, database indexing, and query optimization.

2. Role of the Web Server

- A web server receives requests from students' browsers and sends the appropriate responses.
- Receives HTTP/HTTPS requests from students.
- Processes user requests and application logic.
- Validates student login credentials.
- Communicates with the database server to retrieve or update information.
- Generates or prepares web-page responses and sends them to students.
- Handles multiple requests simultaneously using suitable threads or asynchronous processing.

Example: When a student clicks the “View Results” button, the web server receives the request and asks the database server to retrieve that student's examination results.

3. Role of the Database Server

- A database server stores, organizes, retrieves, and updates the college's information.
- Stores student details and securely stored authentication information.
- Maintains examination results and attendance records.
- Stores course registration and fee-payment details.
- Retrieves information requested by the web server.
- Updates records when students register or make payments.
- Maintains data accuracy, consistency, and security.

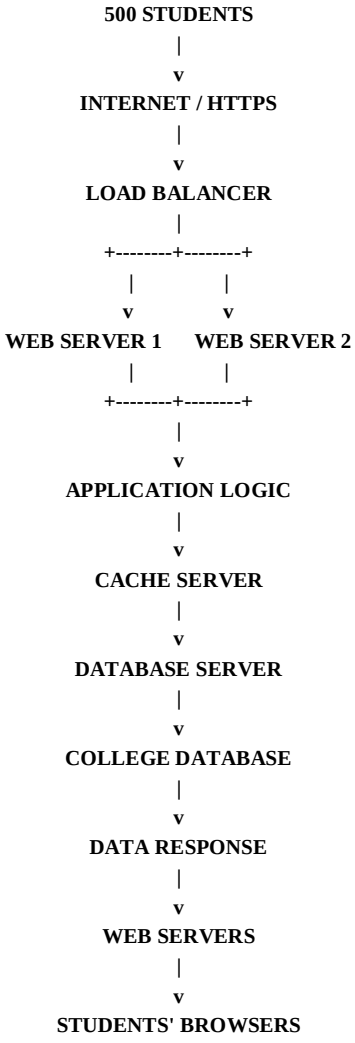
Example: When the web server requests a student's examination results, the database server searches the student records and returns the required information.

4. How the Web Server and Database Server Cooperate

- Student request: Students open the college website using their browsers and send requests.
- Web server receives the request: The web server accepts incoming requests from the students.
- Request processing: The web server validates the request and checks authentication and permissions.
- Database communication: The web server sends an appropriate database query to retrieve or update information.
- Database processing: The database server executes the query and accesses the required records.
- Data response: The database server returns the requested data to the web server.
- Web-page generation: The web server formats the data into a suitable HTML page or JSON response.
- Response to students: The web server sends the response to the students' browsers.

5. Architecture Diagram

The following simplified architecture shows how requests can be distributed and processed:



Note: This is a logical architecture. In a real system, the cache may be placed beside the application servers, and read replicas may be added when needed.

6. Techniques to Handle 500 Simultaneous Users Efficiently

1. Load Balancing

Distributes incoming requests among multiple web servers.

Prevents one server from becoming overloaded.

Improves availability and response time.

Example: A load balancer may distribute 500 requests across two web servers, approximately 250 requests each, depending on their capacity and current load.

2. Caching

Stores frequently accessed information temporarily.

Reduces repeated database queries.

Improves website loading speed.

Example: College announcements and timetables can be cached so that every student does not need to retrieve the same information from the database.

3. Database Indexing

Creates indexes on frequently searched database columns.

Helps retrieve information faster.

Reduces the time needed to execute suitable queries.

Example: An index on the student registration number helps the database find an individual student's results quickly.

4. Connection Pooling

Maintains a group of reusable database connections.

Avoids creating a new connection for every request.

Reduces connection setup overhead and helps control database load.

Example: Instead of opening a new database connection for every student request, the web server reuses available connections from a connection pool.

5. Database and Query Optimization

Uses efficient SQL queries and appropriate database design.

Avoids retrieving unnecessary columns or records.

Uses indexes and limits results where appropriate.

Example: When a student checks attendance, the database retrieves only that student's attendance records instead of retrieving the records of every student.

6. Security and Authentication

Uses HTTPS to protect data in transit.

Validates student login credentials and access permissions.

Uses secure password storage, database permissions, and parameterized queries.

Example: A student should be able to view only their own examination results, not the results of other students.

7. Database Replication and Backup

Maintains additional copies of database information for availability and recovery.

Read replicas can handle suitable read-only requests and reduce load on the primary database.

Regular backups help recover information after failures.

Example: A read replica can handle examination-result lookups while the primary database handles important updates such as fee payments.

7. Practical Example: 500 Students Checking Examination Results

Consider a situation in which 500 students try to check their examination results at the same time.

Step

Server activity

1. Students open the college website.

2 .The load balancer distributes requests among available web servers.

3.The web servers validate student login details and permissions.

4.The application checks whether the required results are available in the cache.

5.If the data is not cached, the application sends a query to the database server or an appropriate read replica.

6.The database retrieves the required examination results.

7.The results are returned to the web server.

8. The web server displays the results in the students' browsers.

Result: All 500 students can access their examination results more efficiently without unnecessarily overloading a single server.

8. Advantages of This Cooperation

High performance: Processes multiple student requests quickly.

Scalability: Additional web servers can be added when student traffic increases.

Reliability: Multiple servers and suitable recovery arrangements reduce the risk of service failure.

Data accuracy: Database transactions help maintain correct and consistent information.

Security: Authentication and access controls protect student records.

Reduced response time: Caching and database optimization improve speed.

Better resource utilization: Load balancing distributes work across available servers.

9. Conclusion

The web server and database server must cooperate to handle 500 simultaneous students efficiently. The web server receives and processes student requests, while the database server stores, retrieves, and updates information. By using load balancing, caching, database indexing, connection pooling, query optimization, and security mechanisms, the college website can provide fast, reliable, and secure services without overloading the servers.

In simple words: The web server acts as a communication bridge between students and the database server, while the database server manages the information. Their proper coordination ensures smooth website performance even when 500 students access it simultaneously.